

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Cyclohexanone oxime

**CAS No** 100-64-1

**Product Code** **L11221**

**Address** ThermoFisher Scientific Australia Pty Ltd  
 5 Caribbean Drive, Scoresby  
 VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292

Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.  
 This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable solids

Category 1

#### Health hazards

Acute Oral Toxicity

Category 4

Serious Eye Damage/Eye Irritation

Category 2

Specific target organ toxicity - (repeated exposure)

Category 2

#### Environmental hazards

Chronic aquatic toxicity

Category 3

### Label Elements



Flame



Exclamation Mark



Health Hazard

**Signal Word****Danger****Hazard Statements**

H302 - Harmful if swallowed

H228 - Flammable solid

H319 - Causes serious eye irritation

H373 - May cause damage to organs through prolonged or repeated exposure

H412 - Harmful to aquatic life with long lasting effects

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P337 + P313 - If eye irritation persists: Get medical advice/attention

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P314 - Get medical advice/attention if you feel unwell

P330 - Rinse mouth

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 - Store in a well-ventilated place

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

**Section 3 - Composition and Information on Ingredients**

| Component           | CAS No   | Weight % |
|---------------------|----------|----------|
| Cyclohexanone oxime | 100-64-1 | <100     |

**Section 4 - First Aid Measures****Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                                                     |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Decomposition Temperature

> 150°C

### Specific Hazards Arising from the Chemical

Flammable. Vapors may form explosive mixture with air. Containers may explode when heated.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Keep away from open flames, hot surfaces and sources of ignition.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

The product does not contain any hazardous materials with occupational exposure limits established.

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Neoprene       | recommendations   |                 |                 |                       |
| Natural rubber |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                      |                                          |
|------------------------------------------------|--------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Beige                                |                                          |
| <b>Physical State</b>                          | Powder Solid                         |                                          |
| <b>Odor</b>                                    | No information available             |                                          |
| <b>Odor Threshold</b>                          | No data available                    |                                          |
| <b>pH</b>                                      | No information available             |                                          |
| <b>Melting Point/Range</b>                     | 88 - 92 °C / 190.4 - 197.6 °F        |                                          |
| <b>Softening Point</b>                         | No data available                    |                                          |
| <b>Boiling Point/Range</b>                     | 206 - 210 °C / 402.8 - 410 °F        | @ 760 mmHg                               |
| <b>Flash Point</b>                             | 90 °C / 194 °F                       | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                       | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available             |                                          |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.3<br><b>Upper</b> 7.8 |                                          |
| <b>Vapor Pressure</b>                          | 0.03 mmHg @ 25 °C                    |                                          |
| <b>Vapor Density</b>                           | Not applicable                       | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available                    |                                          |
| <b>Bulk Density</b>                            | No data available                    |                                          |
| <b>Water Solubility</b>                        | Insoluble                            |                                          |
| <b>Solubility in other solvents</b>            | No information available             |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                      |                                          |
| <b>Component</b>                               | <b>log Pow</b>                       |                                          |
| Cyclohexanone oxime                            | 1.265                                |                                          |
| <b>Autoignition Temperature</b>                | 265 °C / 509 °F                      |                                          |
| <b>Decomposition Temperature</b>               | > 150°C                              |                                          |
| <b>Viscosity</b>                               | Not applicable                       | Solid                                    |
| <b>Explosive Properties</b>                    |                                      | explosive air/vapour mixtures possible   |
| <b>Oxidizing Properties</b>                    | No information available             |                                          |
| <b>Other information</b>                       |                                      |                                          |
| <b>Molecular Formula</b>                       | C6 H11 N O                           |                                          |
| <b>Molecular Weight</b>                        | 113.16                               |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                                         |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                    |
| <b>Conditions to Avoid</b>              | Temperatures above .?1°C, Incompatible products, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids.                                                                             |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                    |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                           |

## Section 11 - Toxicological Information

## Information on Toxicological Effects

## Product Information

## (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component           | LD50 Oral         | LD50 Dermal             | LC50 Inhalation |
|---------------------|-------------------|-------------------------|-----------------|
| Cyclohexanone oxime | 882 mg/kg ( Rat ) | > 5000 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

## (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

## (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

## (g) reproductive toxicity;

Based on available data, the classification criteria are not met

## (h) STOT-single exposure;

Based on available data, the classification criteria are not met

## (i) STOT-repeated exposure;

Category 2

Target Organs

Blood.

## (j) aspiration hazard;

Not applicable

Solid

## Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available

## Section 12 - Ecological Information

## Ecotoxicity effects

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component           | Freshwater Fish                                                    | Water Flea | Freshwater Algae | Microtox                                                                     |
|---------------------|--------------------------------------------------------------------|------------|------------------|------------------------------------------------------------------------------|
| Cyclohexanone oxime | LC50: 189 - 230 mg/L,<br>96h flow-through<br>(Pimephales promelas) |            |                  | EC50 = 19.7 mg/L 5 min<br>EC50 = 22.1 mg/L 15 min<br>EC50 = 25.9 mg/L 30 min |

## Persistence and Degradability

Expected to be biodegradable

## Persistence

Insoluble in water, May persist, based on information available.

Degradation in sewage treatment plant

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative Potential

May have some potential to bioaccumulate

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Cyclohexanone oxime | 1.265   | No data available             |

**Mobility** Spillage unlikely to penetrate soil. The product evaporates slowly. Is not likely mobile in the environment due its low water solubility Spillage unlikely to penetrate soil

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                         |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                     |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains. |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                  |
|--------------------------------|----------------------------------|
| <b>UN-No</b>                   | UN1325                           |
| <b>Proper Shipping Name</b>    | FLAMMABLE SOLID, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b> | Cyclohexanone oxime              |
| <b>Hazard Class</b>            | 4.1                              |
| <b>Packing Group</b>           | III                              |

### ADG

|                                |                                  |
|--------------------------------|----------------------------------|
| <b>UN-No</b>                   | UN1325                           |
| <b>Proper Shipping Name</b>    | FLAMMABLE SOLID, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b> | Cyclohexanone oxime              |
| <b>Hazard Class</b>            | 4.1                              |
| <b>Packing Group</b>           | III                              |

### IATA

|                                |                                  |
|--------------------------------|----------------------------------|
| <b>UN-No</b>                   | UN1325                           |
| <b>Proper Shipping Name</b>    | FLAMMABLE SOLID, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b> | Cyclohexanone oxime              |
| <b>Hazard Class</b>            | 4.1                              |
| <b>Packing Group</b>           | III                              |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

#### Australia

See section 8 for national exposure control parameters.

#### Standard for the Uniform Scheduling of Medicines and Poisons

No poison schedule number allocated.

#### Australian Industrial Chemicals Introduction Scheme (AICIS)

| Component                      | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------------------|-------------------------------------------------------------|------------------------|
| Cyclohexanone oxime - 100-64-1 | Present                                                     | -                      |

#### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

#### National pollutant inventory

Not applicable

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### International Inventories

| Component           | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCs | ISHL | IECSC | KECL       |
|---------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|------------|
| Cyclohexanone oxime | X    | X     | 202-874-0 | -      | X    | X   | -    | X     | X    | X    | X     | KE-19-0001 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

#### Ozone Depletion Potential

This product does not contain any known or suspected substance

#### Persistent Organic Pollutant

This product does not contain any known or suspected substance

#### Rotterdam Convention (PIC)

Not applicable

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component           | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Cyclohexanone oxime | 100-64-1 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

Authorisation/Restrictions according to EU REACH

Not applicable

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
 Predicted No Effect Concentration (PNEC)  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date** 18-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**